

Scalability & Future Plan

Date	10 July 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

The **Credit Card Approval Prediction System** is designed to automate the credit card approval process using machine learning. Although the current implementation works efficiently for local deployment and demonstration, the system can be enhanced to support larger user bases, increased data volumes, stronger security, and cloud-based deployment. This document outlines the current limitations, scalability improvements, and future roadmap for extending the project.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Supports only local deployment using Flask	Cannot be accessed by multiple users over the internet	High
2	Limited dataset size	Prediction accuracy may reduce with unseen or diverse applicant data	Medium
3	Basic security implementation	User authentication and secure API access are not available	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited local users	Deploy on IBM Cloud , AWS , or Azure with load balancing for multiple concurrent users
2	Data Storage	CSV files used for storage	Integrate MySQL , PostgreSQL , or MongoDB for secure and scalable data management
3	Performance	Runs on a local machine	Optimize model, use REST APIs, caching, and cloud computing resources
4	Security	Basic Flask application	Implement user authentication, HTTPS, JWT tokens, role-based access, and encrypted communication

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on IBM Cloud using Watson Machine Learning	1 Month	Enables cloud access and real-time online predictions
Phase 2	Integrate a relational database (MySQL/PostgreSQL)	2 Months	Secure storage and management of applicant information
Phase 3	Develop a mobile application and REST API	3 Months	Improves accessibility for customers and bank employees
Phase 4	Add Explainable AI (XAI), fraud detection, and continuous model retraining	4–6 Months	Increases transparency, improves security, and enhances prediction accuracy